

PHYS 455 Examination

Quantum Mechanics and Wave Functions – Answer Key

1. (B) $\hbar/2$.

The Heisenberg uncertainty principle states $\Delta x \cdot \Delta p \geq \hbar/2$, where $\hbar = h/2\pi$ is the reduced Planck constant. This minimum is achieved for Gaussian wave packets.

2. (B) They scale as n^2 where n is the quantum number.

For the infinite square well, $E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$ where $n = 1, 2, 3, \dots$. The energies are quantized and scale quadratically with n , not evenly spaced.

3. (C) $|\psi(x, t)|^2$ represents the probability density for finding the particle at position x .

The Born rule states that $|\psi(x, t)|^2 dx$ gives the probability of finding the particle in the interval $[x, x + dx]$ at time t . The wave function itself is generally complex-valued.

4. (B) It has real eigenvalues and orthogonal eigenfunctions.

Hermitian operators satisfy $\langle \phi | \hat{A} \psi \rangle = \langle \hat{A} \phi | \psi \rangle$, which guarantees real eigenvalues (measurable quantities) and orthogonal eigenstates (distinct measurement outcomes are distinguishable).

5. *Sample answer:*

The commutator $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ measures whether two observables can be simultaneously measured with arbitrary precision:

- If $[\hat{A}, \hat{B}] = 0$, the observables are compatible: they share common eigenstates and can be simultaneously measured with definite values
- If $[\hat{A}, \hat{B}] \neq 0$, the observables are incompatible: measuring one disturbs the other, and they cannot both have definite values simultaneously
- Example: $[\hat{x}, \hat{p}] = i\hbar$ leads to the uncertainty principle $\Delta x \cdot \Delta p \geq \hbar/2$

This is fundamental to quantum complementarity and distinguishes quantum from classical physics.

6. *Sample answer:*

The quantum harmonic oscillator describes a particle in a quadratic potential $V(x) = \frac{1}{2}m\omega^2 x^2$:

- **Energy spectrum:** $E_n = \hbar\omega(n + 1/2)$ for $n = 0, 1, 2, \dots$. The energies are evenly spaced, and the ground state has non-zero energy $E_0 = \hbar\omega/2$ (zero-point energy)
- **Wave functions:** Involve Hermite polynomials and Gaussians, symmetric (even n) or antisymmetric (odd n)
- **Significance:** Models molecular vibrations, phonons in solids, quantum field theory modes, and electromagnetic field oscillations. It's exactly solvable and exhibits key quantum features like quantization and zero-point motion

7. *Sample answer:*

Time-dependent Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H} \psi$$

This governs the time evolution of quantum states and is the equation of motion analogous to Newton's second law in classical mechanics.

Time-independent version: When the Hamiltonian is time-independent, we can separate variables: $\psi(x, t) = \phi(x)e^{-iEt/\hbar}$, leading to:

$$\hat{H}\phi = E\phi$$

This eigenvalue equation determines stationary states (energy eigenstates).

When to use each:

- Time-independent: Finding energy eigenstates, bound state problems, stationary potentials
- Time-dependent: Evolution of superpositions, time-varying potentials, scattering dynamics

8. *Sample answer:*

Quantum tunneling: A quantum particle can penetrate and traverse potential barriers even when its energy E is less than the barrier height V_0 (classically forbidden).

Wave nature explanation:

- The wave function doesn't abruptly vanish at a barrier; it decays exponentially inside the barrier region as $\psi \sim e^{-\kappa x}$ where $\kappa = \sqrt{2m(V_0 - E)}/\hbar$
- If the barrier is finite in width, the wave function has non-zero amplitude on the other side
- This leads to non-zero transmission probability even for $E < V_0$

Implications: Tunneling enables radioactive alpha decay, scanning tunneling microscopy (STM), tunnel diodes, and is essential in nuclear fusion in stars. It has no classical analog and demonstrates the wave-particle duality of quantum mechanics.